Cone Spacing Guide

Temporary Traffic Management Tapers and Delineation

1. Introduction

This guide explains the cone spacing requirements for temporary traffic management, including the reasoning behind different spacing requirements for various taper types. The information is sourced from the Austroads Guide to Temporary Traffic Management (AGTTM) Part 3: Static Worksites.

Understanding cone spacing is essential for:

- Ensuring road worker safety
- Providing clear guidance to road users
- Meeting regulatory requirements
- Reducing the risk of crashes at worksites

2. Cone Spacing Requirements (Table 5.3)

The following table shows the maximum cone spacing for traffic cones and bollards based on speed and purpose. These are maximum measurements and can be reduced for curves, night works, or where there is a risk road users may take a wrong turn.

Purpose	Speed	Max Spacing
All purposes	≤ 55 km/h	4 m
All purposes	56 – 75 km/h	12 m
All purposes	≥ 76 km/h	18 m
Protecting freshly painted lines	56 – 75 km/h	24 m
Protecting freshly painted lines	≥ 76 km/h	60 m
Taper at traffic control station	All speeds	4 m
Centreline approach to traffic controller	All speeds	4 m
Crossover for contraflow	All speeds	2 m

Table 1: Cone Spacing Requirements (AGTTM Part 3, Table 5.3)

3. Taper Types and Lengths (Table 5.7)

Different taper types serve different purposes and require different lengths. The taper length determines how far in advance traffic control devices must be placed to safely guide road users through the transition.

Speed	Traffic Control Taper	Lateral Shift Taper	Merge Taper
≤ 45 km/h	15 m	5 m	15 m
46 – 55 km/h	15 m	15 m	30 m
56 – 65 km/h	30 m	30 m	60 m
66 – 75 km/h	N/A	70 m	115 m
76 – 85 km/h	N/A	80 m	130 m
86 – 95 km/h	N/A	90 m	145 m
≥ 96 km/h	N/A	100+ m	160+ m

Table 2: Recommended Taper Lengths (AGTTM Part 3, Table 5.7)

4. Taper Types Explained

4.1 Traffic Control Taper

Purpose: Used where there is a Portable Traffic Control Device (PTCD) or traffic controller prior to a lateral shift. Traffic is expected to STOP at this location.

Characteristics:

• Only used at speeds up to 65 km/h

• Always uses 4 m cone spacing regardless of speed

• Includes 4 lead-in cones at 4 m spacing on centreline

• Traffic controller positioned 6 m in front of taper

Why tight cone spacing? The tight 4 m spacing serves multiple safety purposes:

• Visually narrows the road, signalling drivers to slow down and stop

• Protects the traffic controller who is stationed at this location

• Prevents vehicles from bypassing the control point

• Provides clear delineation for queued vehicles

4.2 Lateral Shift Taper

Purpose: Used where traffic is required to shift laterally without closing another traffic stream. Traffic KEEPS MOVING through the taper.

Characteristics:

• Used at all speed ranges

• Cone spacing based on speed (Table 5.3)

• Taper length based on speed (Table 5.7)

• No person stationed at the taper

Why longer taper with wider spacing?

• Vehicles maintain speed through the shift

• Longer taper allows smooth lateral transition

• No worker protection required at taper location

4.3 Merge Taper

Purpose: Used on multilane roads where one lane of traffic merges into another lane. Traffic CONTINUES FLOWING while merging.

Characteristics:

• Longest taper type (approximately 1.5× lateral shift taper length)

• Cone spacing based on speed (Table 5.3)

• Merge occurs one lane at a time

• No lead-in cones required

Why longest taper?

• Merging requires complex driver decision-making

• Drivers must judge gaps, adjust speed, and yield

• Extended length provides more time for merging manoeuvre

• Continuous flow maintained throughout

5. Why Different Cone Densities?

A common question is why traffic control tapers have more cones over a shorter distance compared to merge tapers with fewer cones over a longer distance. The answer lies in the purpose of each taper type.

Factor	Traffic Control Taper	Merge Taper
Traffic action	STOP	CONTINUE/MERGE
Person at taper	Yes (traffic controller)	No
Cone spacing	4 m (tight)	12-18 m (wider)
Taper length (60 km/h)	30 m	60 m
Number of cones	~8 cones	~5 cones
Primary function	Protect worker, signal stop	Guide merge at speed

Table 3: Comparison of Taper Types at 60 km/h

Key Insight: Cone Spacing as a Speed Control Treatment

The AGTTM specifically identifies close spacing of delineation devices as a treatment to reduce traffic speeds. Tight cone spacing visually narrows the road and signals to drivers that they must slow down. This is why traffic control tapers, where vehicles must stop, use the tightest 4 m spacing regardless of speed.

6. Practical Example: 60 km/h Worksites

Consider two worksites on a road with a 60 km/h speed limit:

Scenario 1: Traffic Control Taper (Stop/Slow Operation)

Element	Specification	Purpose
Lead-in cones	4 cones @ 4 m (16 m)	Prevent bypass, signal stop
Taper length	30 m	Shorter - vehicles stopping
Cone spacing	4 m	Tight - protect controller
Cones in taper	8 cones	High density visual cue
Total cones	12 cones	-

Scenario 2: Merge Taper (Lane Closure)

Element	Specification	Purpose
Lead-in cones	None	Not required for merge
Taper length	60 m	Longer - continuous flow
Cone spacing	12 m	Wider - moving traffic

Cones in taper	5 cones	Guide at speed
Total cones	5 cones	-

7. Summary

The key principles for understanding cone spacing in temporary traffic management are:

1. Speed determines base spacing — Higher speeds allow wider spacing (4 m at ≤55 km/h, 12 m at 56-75 km/h, 18 m at ≥76 km/h).

2. Purpose overrides speed — Special purposes like traffic control stations require 4 m spacing regardless of speed.

3. Taper type reflects function — Traffic control tapers are short with tight spacing (stop location); merge tapers are long with wider spacing (continuous flow).

4. Tight spacing = speed reduction — Close cone spacing is itself a treatment to reduce traffic speeds and signal to drivers that they need to slow down or stop.

5. Worker protection is paramount — Where traffic controllers are positioned, tighter spacing provides both physical protection and visual signalling to drivers.

Quick Reference

Situation	Cone Spacing
Traffic control station (any speed)	4 m
General taper at ≤55 km/h	4 m
General taper at 56-75 km/h	12 m
General taper at ≥76 km/h	18 m
Contraflow crossover	2 m

Reference

Austroads (2021). Guide to Temporary Traffic Management, Part 3: Static Worksites. Tables 5.3 and 5.7, Pages 64 and 77.